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AN OVERVIEW OF THE TIME-RESOLVED CAPABILITIES AND SAMPLE SETUP MODULARITY AT CoSAXS

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Abstract

CoSAXS is a multipurpose SAXS instrument located at the 3 GeV ring of MAX IV Laboratory in Sweden. This instrument provides a versatile platform for conducting Small-Angle X-ray scattering (SAXS) experiments on a wide range of research fields. With an extensive pool of sample environments, CoSAXS enables the application of multiple techniques and complex experiments on solid and solution samples.

To accommodate the high demand and facilitate the rapid exchange of sample setups, a standardized mounting system has been implemented, and additive manufacturing techniques are utilized for fast and efficient prototyping and production of customized sample holders. Furthermore, CoSAXS is equipped with advanced sample environments, such as the setup for milliseconds time-resolved SAXS-WAXS experiments in solution (TR-XSS). Among other studies, it has been used in non-reversible protein reactions after laser activation of caged compounds.

OVERVIEW OF CoSAXS CAPABILITIES

CoSAXS is a state-of-the-art multipurpose instrument located at the 3 GeV ring of the MAX IV Laboratory.

A 2 meter long in-vacuum undulator, coupled with a Si (111) HDCM and a Dual KB mirror pair, provides an estimated photon flux of 10^{12} - 10^{13} ph/s. The beam energy can be varied in the range between 4 and 20 keV, with the possibility of changing the focus position, so that the beam dimension at the sample position can vary between $25 \times 15\ \mu m^2$ and $150 \times 150\ \mu m^2$.

The SAXS/WAXS data can be detected and recorded simultaneously using dedicated SAXS and custom L-shaped WAXS detectors installed in a 16 m long vacuum vessel. In addition, a 1-D detector can be mounted on the outside of the vessel.

The detector configuration makes possible the collection of continous data in the q-range between 6×10^{-4} and $3\ Å^{-1}$ [1].

These specifications make CoSAXS a versatile instrument for different kinds of experiment on both solid-state and solution samples.

The pool of sample setups available at CoSAXS includes for example, a multiple capillary holder, a pipetting robot for samples in solution, Linkam heating stages, cells for microfluidics, a rheometer, a stopped flow device, a high magnetic field electromagnet, an ultrasonic levitator, a biaxial mechanical tester, a setup for combined SAXS with UV–vis and fluorescence spectroscopy (SUrF) [2, 3].

The beamline also includes a specialized time-resolved setup based on laser excitation in the millisecond range. This setup will be discussed in detail in this paper.

Most of the setups are mounted on an x-y stage stack with micrometric precision, for precise alignment and scanning of the sample position, and both beamline and user defined equipment can be mounted.

TIME RESOLVED CAPABILITIES

Time-resolved X-ray solution scattering (TR-XSS) is a technique used to monitor the structural changes in a solution with time, after an initial perturbation and over a variety of time scales.

At CoSAXS, a nanosecond laser with an optical parametric oscillator (OPO) tunable from 675 to 2500 nm (20–140 mJ) and a continuous infrared laser (1470 nm, 50 J energy) are available as triggering for the experiments. These lasers are fiber optically coupled, to transport the light to the sample environment. The fiber optics simplifies the alignment process, guarantees consistent light delivery, significantly reduces the hazards and mitigates the stringent safety requirements typically associated with open beam paths. Other excitation sources, like diode-based monochromatic sources on the UV-VIS-Infrared regime, could be employed in the future.

In one study at CoSAXS, TR-XSS has been exploited to monitor the changes in an enzyme when a caged compound was activated by a nanosecond UV laser pulse. The reaction was monitored up to 50 ms, with a 2 ms temporal resolution [4, 5].

The core of the experimental setup, shown in Fig. 1, revolves around a flow-through cell, designed to accommodate a quartz capillary. Initially, this cell was fabricated from an aluminum block, featuring a precisely machined hole to host 1.5 mm diameter capillaries [6]. However, due to experimental requirements, the demand for greater flexibility in capillary sizes became apparent. To address this, the design was shifted to a 3D-printed capillary block, which offers significant advantages, primarily the ability to rapidly, and cost-effectively, cater for this demand. Given that the system can operate at temperatures ranging from few degrees up to 75 °C, PETG was selected as the ideal material for 3D printing due to its thermal and mechanical resistance.

At the heart of the block lies a central opening. This aperture is where the X-ray beam and the excitation light, perpendicular to each other, converge, allowing them to interact precisely with the same sample volume within the capillary.

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A dedicated sample viewer with micrometric resolution is used, to facilitate the laser/X-ray alignment [7] and a photodiode monitors the laser output intensity.

To maintain a stable temperature within the flow-through cell, and consequently inside the capillary, the central block is sandwiched between two copper blocks equipped with integrated water channels, which are connected to a water circulation bath. This bath provides precise temperature regulation, capable of maintaining the sample anywhere between 5 °C and 75 °C, ensuring optimal experimental conditions.

A 10 µm wall thickness, uniform diameter capillary, open at both ends, is carefully inserted into the central block. The sample solution is contained in an Eppendorf tube placed under the capillary. Less than 300 µL are used for a typical TR-XSS experiment. PVC tubing and a peristaltic pump create a closed-loop system, continuously circulating the sample between the Eppendorf tube and the capillary.

One issue often associated with X-ray exposure in high brilliant sources is the radiation damage in sensitive samples, which frequently manifests itself as aggregation of the samples on the capillary wall. To overcome this problem, the capillary assembly is connected to a motorized micrometric screw, that can move the capillary in very small steps, in order to expose a new, clean section of the capillary.

A schematic of the system is presented in Fig. 1 and a picture of the setup at CoSAXS shown in Fig. 2.

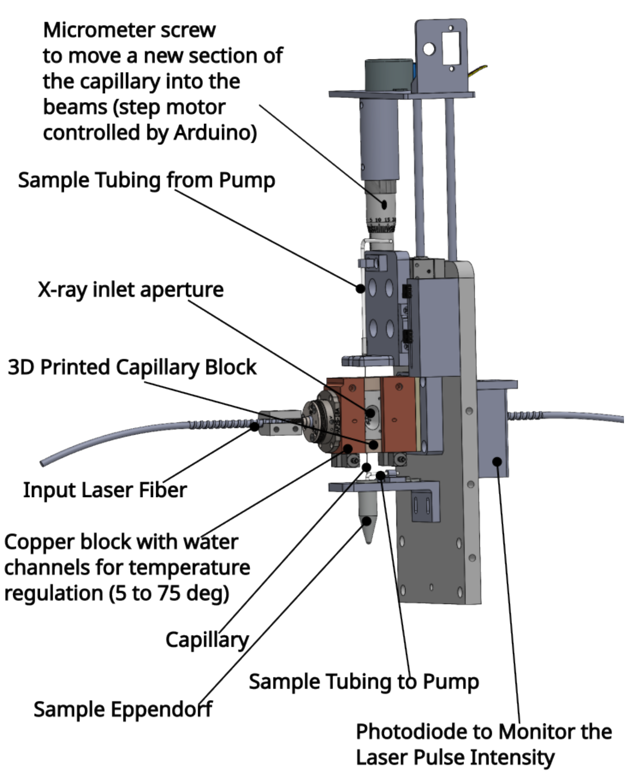

Figure 1: CAD model of the TR-XSS setup. The main components are indicated.

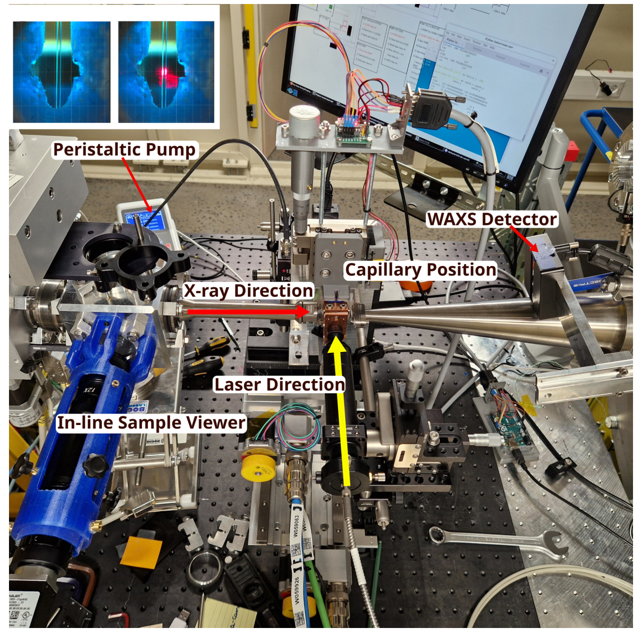

Figure 2: A picture of the experimental setup installed at the CoSAXS beamline, showing the direction of the laser and X-ray beams and highlighting the main components. An inset in the upper-left corner displays a shot from the in-line sample viewer, comparing the capillary image with the exciting laser light off and on.

SAMPLE HOLDER DIVERSITY AND STANDARDIZATION

A wide range of experiments is performed daily at CoSAXS, introducing the need to accommodate the high demand and facilitate the rapid exchange of sample setups.

A standardized mounting system has been implemented and mounted on a stack of X-Y motorized stages (perpendicular to the beam) and a manual Z-stage (in the beam direction). The X-Y stages have absolute linear encoders that provide a position reproducibility of 1 µm. The advantage of a precise X-Y-Z stack is the ability to always mount the sample holder in the same position, guaranteeing the sample-to-detector distance and minimizing the need for recalibration.

A large variety of sample holders can be mounted on the stages. A few examples are shown in Fig. 3. In particular, picture 3B is a CAD drawing of a 3D printed holder for cuvettes, while Fig. 3C shows an holder for solid samples with multiple holes and different sizes; the plates can be easily mounted and unmounted, and multiple plates are available for the users, to replace the samples off-line.

A sliding base has been developed for sample holders that require frequent mounting and un-mounting during a beamtime for sample exchange. This base is shown in Fig. 3A.

One example of such sliding sample holders is reported in Fig. 3D. This is an aluminum holder hosting a flow-through cell, manufactured using Powder Bed Fusion (PBF) additive manufacturing. This cell has been used for example during experiments where an ex-situ reaction is monitored in specific times, injecting the samples into the capillary and

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the X-ray path [8]. The sample holder shown in picture 3B was employed during the investigation of the aqueous liquid suspension of high aspect ratio 2D clay nanosheets [9].

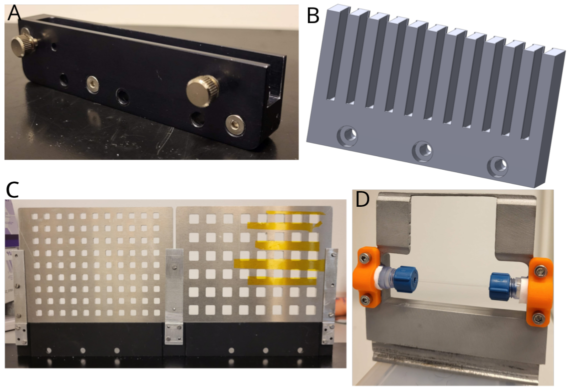

Figure 3: Examples of Sample holders mounted on the Z-translation stage: (A) Sliding base allowing quick and reproducible sample holder positioning.(B) 3D printed holder for cuvettes [9].(C) Frames for multiple samples (solid and gels).(D) Aluminum holder created using PBF (Powder Bed Fusion) with space for a flow-through cell with quick connectors (bottom) and multiple disposable capillaries on the top [8].

3D PRINTED METAL MULTICAPILLARY HOLDER

At CoSAXS, a common experiment involves analyzing samples in solution within disposable borosilicate or quartz capillaries with thin walls. To maximize the number of capillaries scanned during an experiment and to improve the beamline efficiency, a multicapillary holder has been developed, capable of holding up to 26 capillaries, each 1.5 mm in diameter (Fig. 4).

The holder easily mounts and unmounts from the X-Y-Z stage system via a sliding base, which enables safe and convenient off-line filling. To further enhance efficiency, CoSAXS provides two such holders that can be swapped: one for off-line preparation and the other for mounting on the beamline.

For certain solution-based experiments, temperature stability of ±0.1 °C is critical. For instance, many drug delivery systems require structural information at body temperature, while other experiments may necessitate data collection across a range of temperatures, or during temperature ramps.

To address this, the multicapillary holder was designed and manufactured with an integrated, hollow channel for circulating temperature-controlled water using additive manufacturing (aluminum Metal Laser Sintering), see Fig. 4. Traditional manufacturing methods were unsuitable due to the risk of leaks from multi-part construction.

A water circulation bath controls the water temperature between 5 °C and 75 °C. To test and calibrate the system's temperature stability, thermocouples were placed at various points on the holder and inside water-filled capillaries. The control of the water bath and the readout of the thermocouples temperature is integrated in the Tango/Sardana based control system, and these values are recorded in the experiment metadata (see Ref. [5], for further information regarding the CoSAXS control system).

As shown in Fig. 5 the system demonstrates excellent homogeneity and stability, with most thermocouples closely tracking the circulating water bath within a range of 10 °C to 75 °C.

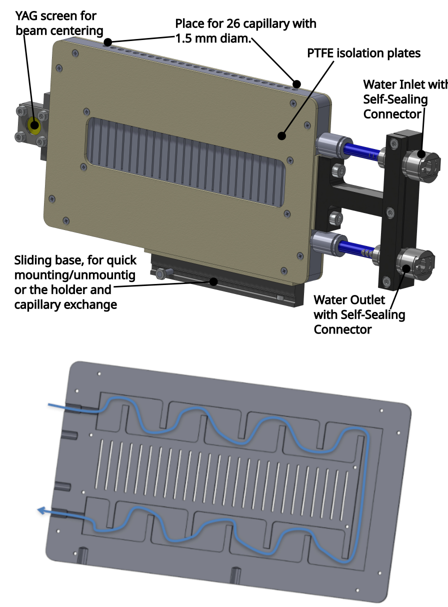

Figure 4: The 3D-printed aluminum multicapillary holder. The sectioned view at the bottom of the figure highlights, in blue, the integrated channel for cooling water.

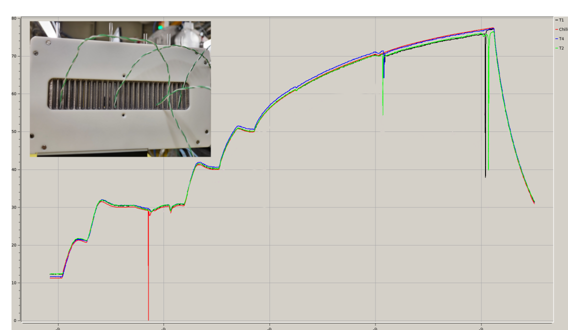

Figure 5: Temperature calibration of the multicapillary holder. The graph shows the chiller set temperature (red line) and the temperature readings from multiple thermocouples placed on the holder (remaining lines). The thermocouples closely track the circulating bath temperature across the 10 °C and 75 °C range, demonstrating excellent stability.

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